

Quantum Channels as Noncommutative Markov Operators

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Abstract

This project studies quantum channels as noncommutative analogues of classical Markov operators. We first introduce Markov kernels and their associated operators on spaces of measurable functions, emphasizing positivity, unitality and normality. We then move to the noncommutative setting of bounded operators on a Hilbert space and introduce quantum channels as normal completely positive unital maps. Finally, we present Kraus representations and show, through the restriction to diagonal operators, how classical Markov operators arise from quantum channels.

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1 Introduction

Markov transition kernels describe stochastic evolutions on measurable spaces and induce positive unital operators on spaces of measurable functions through integral representations. In classical probability theory, these operators provide an operator-theoretic description of Markovian dynamics.

In quantum theory, states are represented by density operators on a Hilbert space and their evolutions by quantum channels, namely completely positive and trace-preserving maps. Passing to the dual picture leads naturally to normal completely positive unital maps on von Neumann algebras of bounded operators.

The purpose of this project is to study quantum channels as noncommutative analogues of classical Markov operators. We first develop the measure-theoretic and operator-theoretic framework of Markov kernels and their associated operators, together with their duality and normality properties. We then introduce quantum channels and their Kraus representations, highlighting the parallels between classical stochastic dynamics and quantum open-system evolutions.

Finally, we show that restricting certain quantum channels to diagonal operators recovers classical Markov operators associated with stochastic matrices and Markov kernels.

2 Markov Operators and Kernels

2.1 Measure-Theoretic and Operator-Algebraic Preliminaries

We begin by recalling the basic measure-theoretic framework and important examples.

Definition 2.1. Let X be a set. A σ -algebra $\mathcal{A}$ on X is a collection of subsets of X satisfying:

- $X \in \mathcal{A}$,
- if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$,
- if $(A_n)_{n \in \mathbb{N}} \subset \mathcal{A}$, then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{A}$.

The pair $(X, \mathcal{A})$ is called a *measurable space*.

Example 2.2. Let $\mathcal{F} \subset 2^X$. Then

$$\sigma(\mathcal{F}) := \bigcap_{\substack{\mathcal{G} \text{ } \sigma\text{-algebra} \\ \mathcal{F} \subset \mathcal{G}}} \mathcal{G}$$

is a σ -algebra, called the σ -algebra generated by $\mathcal{F}$.

The σ -algebra generated by the open sets of $\mathbb{R}^n$ is called the *Borel σ -algebra*, denoted $\mathcal{B}(\mathbb{R}^n)$. Similarly, $\mathcal{B}([0, +\infty])$ is generated by all sets of the form $[a, +\infty]$ with a in $\mathbb{Q}^+$ or $\mathbb{R}^+ [10]$.

Definition 2.3. A *measure* on a measurable space $(X, \mathcal{A})$ is a map $\mu : \mathcal{A} \rightarrow [0, \infty]$ such that:

- $\mu(\emptyset) = 0$,
- for every sequence $(A_n)_{n \in \mathbb{N}}$ of pairwise disjoint sets in $\mathcal{A}$,

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n).$$

The triple $(X, \mathcal{A}, \mu)$ is called a *measure space*.

A measure μ such that $\mu(X) = 1$ is called a *probability measure* and $(X, \mathcal{A}, \mu)$ is then a *probability space*.

Example 2.4. Let $(X, \mathcal{A})$ be a measurable space and $x \in X$. The map

$$\delta_x(A) := \mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{otherwise,} \end{cases}$$

defines a probability measure on $(X, \mathcal{A})$ called the *Dirac measure/mass at x* [10].

Definition 2.5. Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. A map $S : X \rightarrow Y$ is said to be $\mathcal{A}/\mathcal{B}$ -measurable if

$$S^{-1}(B) \in \mathcal{A} \quad \text{for all } B \in \mathcal{B}.$$

If $Y = \mathbb{R}$ and $\mathcal{B} = \mathcal{B}(\mathbb{R})$, then S is simply called a *measurable function*.

We now recall some definitions from C^* -algebras following [9] and [6].

Definition 2.6. An (associative) *algebra* $\mathfrak{A}$ over the complex field $\mathbb{C}$ is a vector space endowed with a product $(a, b) \in \mathfrak{A} \times \mathfrak{A} \mapsto ab \in \mathfrak{A}$ such that :

1. $a(b + c) = ab + ac$,
2. $(a + b)c = ac + bc$,
3. $a(\beta b) = \beta ab = (\beta a)b$,
4. $a(bc) = (ab)c$,

for all $a, b, c \in \mathfrak{A}, \beta \in \mathbb{C}$.

Definition 2.7. A $*$ -*algebra* is an algebra $\mathfrak{A}$ over the complex field $\mathbb{C}$ together with a map $*$: $\mathfrak{A} \rightarrow \mathfrak{A}$ such that :

1. $(a^*)^* = a$,
2. $(ab)^* = b^*a^*$,
3. $(\alpha a + \beta b)^* = \bar{\alpha}a^* + \bar{\beta}b^*$,

for all $a, b \in \mathfrak{A}$ and $\alpha, \beta \in \mathbb{C}$.

1. states that $*$ is an involution. Elements of $\mathfrak{A}$ the form $a = \sum_{k=1}^n b_k^* b_k$, $n \in \mathbb{N}$, $b_k \in \mathfrak{A}$, are called *positive* and form the cone of positive elements denoted by $\mathfrak{A}^+$, which introduces a partial order on the set of self adjoint elements $\mathfrak{A}_{sa} = \{h \in \mathfrak{A} : h^* = h\}$ by

$$a \leq b \iff b - a \in \mathfrak{A}^+ \text{ and } a, b \in \mathfrak{A}_{sa}.$$

Definition 2.8. A $*$ -algebra $\mathfrak{A}$ endowed with a norm $\|\cdot\|$ is a *Banach $*$ -algebra* if it is complete in the metric induced by the norm, the involution is isometric, i.e. $\|a\| = \|a^*\|$ for all $a \in \mathfrak{A}$, and $\|ab\| \leq \|a\|\|b\|$ for all $a, b \in \mathfrak{A}$.

A Banach $*$ -algebra is a *C*-algebra* if moreover $\|a^*a\| = \|a^*\|\|a\| = \|a\|^2$ for all $a \in \mathfrak{A}$.

Remark 2.9. If $\mathfrak{A}$ is a C*-algebra then $\mathfrak{A}^+ = \{b^*b : b \in \mathfrak{A}\}$ by Proposition II.3.1.3 in [4].

Example 2.10. Let (X, Ω) be a measurable space. Then the space $\mathcal{B}(X, \Omega)$ of all bounded complex-valued measurable functions endowed with the uniform norm $\|g\|_\infty = \sup_{x \in X} |g(x)|$ is a commutative C^* -algebra for the involution given by complex conjugation. Indeed it is well known that $\ell^\infty(X)$, the space of all bounded complex valued functions on X , is a commutative C^* -algebra (w.r.t. $\|\cdot\|_\infty$). It is easy to see that $\mathcal{B}(X, \Omega)$ is a norm-closed $*$ -subalgebra of $\ell^\infty(X)$.

Moreover, recall that a positive element in $\mathcal{B}(X, \Omega)$ is of the form $g = \bar{f}f = |f|^2$ for some $f \in \mathcal{B}(X, \Omega)$, i.e., $\mathcal{B}(X, \Omega)^+$ consists of non-negative real-valued functions.

Definition 2.11. Let $\mathfrak{A}, \mathfrak{B}$ be two $*$ -algebras. A linear map $\Phi : \mathfrak{A} \rightarrow \mathfrak{B}$ is *positive* if $\Phi(\mathfrak{A}^+) \subset \mathfrak{B}^+$.

Remark 2.12. Because $a^*a \in \mathfrak{A}^+$, for any $a \in \mathfrak{A}$, and Φ is linear, it suffices to check that $\Phi(a^*a) \in \mathfrak{B}^+$ for all $a \in \mathfrak{A}$.

Definition 2.13. Let $\mathfrak{A}, \mathfrak{B}$ be two $*$ -algebras. A linear map $\Phi : \mathfrak{A} \rightarrow \mathfrak{B}$ is *completely positive* if for any finite collections $a_1, \dots, a_n \in \mathfrak{A}$ and $b_1, \dots, b_n \in \mathfrak{B}$, the element

$$\sum_{i,j=1}^n b_i^* \Phi(a_i^* a_j) b_j \in \mathfrak{B}$$

is positive.

2.2 Markov Kernels

We start by introducing Markov kernels, which are the building blocks of Markov operators.

Definition 2.14 (Markov kernel). Let $(S, \mathcal{S})$ and $(T, \mathcal{T})$ be measurable spaces. A map $\kappa : S \times \mathcal{T} \rightarrow [0, 1]$ is a *Markov kernel* from S to T if

1. for all $B \in \mathcal{T}$, the mapping $s \mapsto \kappa(s, B)$ is $\mathcal{S}$ -measurable,
2. for all $s \in S$, the mapping $B \mapsto \kappa(s, B)$ is a probability measure on $(T, \mathcal{T})$.

Example 2.15. Using Example 2.4, we have that for any measurable space $(X, \mathcal{A})$, the set of Dirac measures $\delta_x, x \in X$ can be seen as a Markov kernel [5] from X to X in the following way :

$$\begin{aligned} \delta : X \times \mathcal{A} &\rightarrow [0, 1] \\ (x, A) &\mapsto \delta_x(A) \end{aligned}$$

When S and T are countable sets, $\mathcal{S} = 2^S$ and $\mathcal{T} = 2^T$, κ is determined by the quantities $K_{ij} = \kappa(i, \{j\})$ for $i \in S$ and $j \in T$ (indeed, $\kappa(i, B) = \sum_{j \in B} \kappa(i, \{j\})$, $\forall B \in \mathcal{T}$).

These numbers define an infinite row stochastic matrix (K_{ij}) , i.e. $K_{ij} \geq 0$, $\forall i \in S \forall j \in T$ and $\sum_{j \in T} K_{ij} = 1$, $\forall i \in S$.

2.3 Markov Operators and Integral Representations

Markov operators have different definitions in the literature, which will lead us to different interpretations. Here's the first one :

Definition 2.16 (Markov operator on bounded measurable functions). Let (X, Ω) and (Y, Σ) be measurable spaces.

Any linear map $T : \mathcal{B}(Y, \Sigma) \rightarrow \mathcal{B}(X, \Omega)$ that satisfies the following conditions is called a *Markov operator* (on bounded measurable functions) :

1. $Tg \geq 0$ for all $g \geq 0$ (positivity),
2. $T\mathbf{1}_Y = \mathbf{1}_X$ (unital).

One can associate Markov operators to Markov kernels as follows :

Let $P : X \times \Sigma \rightarrow [0, 1]$ be a Markov kernel from X to Y and define the map $T_P : \mathcal{B}(Y, \Sigma) \rightarrow \mathcal{B}(X, \Omega)$ by

$$(T_P g)(x) = \int_Y g(y) P(x, dy),$$

for every $g \in \mathcal{B}(Y, \Sigma), x \in X$.

This definition makes sense since g is integrable as a bounded function w.r.t. to a finite

measure and $\sup_{x \in X} |(T_P g)(x)| \leq \sup_{y \in Y} |g(y)|$, i.e. $\|T_P g\|_\infty \leq \|g\|_\infty$.

Then T_P is a Markov operator. Indeed, T_P is clearly linear,

$$T_P \mathbf{1}_Y = \int_Y P(x, dy) = P(x, Y) = \mathbf{1}_X$$

and $g \geq 0 \implies T_P g \geq 0$.

Moreover, we notice that T_P satisfies a stronger property than positivity.

Proposition 2.17. *T_P is completely positive.*

Proof. For any $x \in X$ and any finite collection of elements $f_i \in \mathcal{B}(X, \Omega)$ and $g_i \in \mathcal{B}(Y, \Sigma)$ ($i = 1, \dots, n$), we have

$$\begin{aligned} \sum_{i,j=1}^n f_i(x) T_P(g_i \bar{g}_j)(x) \overline{f_j(x)} &= \sum_{i,j=1}^n f_i(x) \overline{f_j(x)} \int_Y g_i(y) \overline{g_j(y)} P(x, dy) \\ &= \int_Y \sum_{i,j=1}^n f_i(x) g_i(y) \overline{f_j(x) g_j(y)} P(x, dy) \\ &= \int_Y \left(\sum_{i=1}^n f_i(x) g_i(y) \right) \overline{\left(\sum_{j=1}^n f_j(x) g_j(y) \right)} P(x, dy) \\ &= \int_Y \left| \sum_{i=1}^n f_i(x) g_i(y) \right|^2 P(x, dy) \geq 0 \end{aligned}$$

□

We now provide a continuity property for T_P that will later make sense in the context of normality. We first recall that a sequence $\{g_n\}_{n \in \mathbb{N}}$ in $\mathcal{B}(Y, \Sigma)$ is said to be bounded if there exists some $M > 0$ such that $\|g_n\|_\infty \leq M$, $\forall n \in \mathbb{N}$.

Proposition 2.18 (Sequential normality). *For any bounded, increasing sequence $\{g_n\}_{n \in \mathbb{N}}$ of functions in $\mathcal{B}(Y, \Sigma)^+$, we have*

$$T_P(\sup_n g_n) = \sup_n T_P(g_n)$$

Proof. For every $y \in Y$, the sequence $\{g_n(y)\}_{n \in \mathbb{N}}$ is bounded and increasing in $\mathbb{R}$ thus $\lim_{n \rightarrow \infty} g_n(y) = \sup_n g_n(y)$ exists at every $y \in Y$. Similarly, for all $x \in X$, the sequence $\{\int_Y g_n P(x, dy)\}_{n \in \mathbb{N}}$ is increasing and bounded in $\mathbb{R}$, since we are integrating w.r.t. to a finite and positive measure.

Now, $\{g_n\}_{n \in \mathbb{N}}$ bounded $\implies |g_n(y)| \leq \|g_n\|_\infty \leq M \forall n \in \mathbb{N}, \forall y \in Y$. Moreover, the function $M \mathbf{1}_Y$ is itself a bounded function hence is integrable w.r.t. to the finite measure $P(x, \cdot)$ for each $x \in X$. We obtain

$$\begin{aligned} \left[T_P(\sup_n g_n) \right] (x) &= \int_Y \sup_n g_n P(x, dy) \\ &= \int_Y \lim_{n \rightarrow \infty} g_n P(x, dy) \\ &= \lim_{n \rightarrow \infty} \int_Y g_n P(x, dy) \\ &= \sup_n \int_Y g_n P(x, dy) \\ &= \left[\sup_n T_P(g_n) \right] (x) \end{aligned}$$

for all $x \in X$, by application of the Dominated Convergence Theorem. $\square$

Moreover, this property allows us to create a correspondence between Markov kernels and operators :

Proposition 2.19. *Let $T : \mathcal{B}(Y, \Sigma) \rightarrow \mathcal{B}(X, \Omega)$ be a sequentially normal Markov operator and define $P : X \times \Sigma \rightarrow [0, 1]$ by :*

$$P(x, A) := (T\mathbf{1}_A)(x)$$

for all $x \in X, A \in \Sigma$.

Then P is a Markov kernel from X to Y such that $T = T_P$.

We will call T_P the *integral representation* of T via Markov kernels.

Proof. P is well defined : $\mathbf{1}_A \geq 0 \implies (T\mathbf{1}_A)(x) \geq 0$ and

$$\mathbf{1}_Y - \mathbf{1}_A \geq 0 \implies T(\mathbf{1}_Y - \mathbf{1}_A) \geq 0 \implies (T\mathbf{1}_Y)(x) \geq (T\mathbf{1}_A)(x) \implies (T\mathbf{1}_A)(x) \leq 1,$$

hence $0 \leq P(x, A) \leq 1$ for all $x \in X$ and $A \in \Sigma$.

P is a Markov kernel :

1. Fix $A \in \Sigma$. Then $T\mathbf{1}_A \in \mathcal{B}(X, \Omega) \implies T\mathbf{1}_A$ is Ω -measurable, i.e. the mapping $x \mapsto P(x, A) = (T\mathbf{1}_A)(x)$ is Ω -measurable.
2. Fix $x \in X$. Then $P(x, \emptyset) = (T\mathbf{1}_\emptyset)(x) = (T\mathbf{0})(x) = 0$ because T is linear, $P(x, Y) = (T\mathbf{1}_Y)(x) = \mathbf{1}_X(x) = 1$ because T is unital.

Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of pairwise disjoint sets in Σ and define $g_n = \sum_{k=1}^n \mathbf{1}_{A_k}$

for every $n \in \mathbb{N}$. For all $y \in Y$, $|g_n(y)| \leq 1$, since y belongs to at most one the disjoint sets, hence the sequence $\{g_n\}_{n \in \mathbb{N}}$ is bounded and increasing in $\mathcal{B}(Y, \Sigma)^+$.

We obtain

$$\begin{aligned} P(x, \bigcup_n A_n) &= (T\mathbf{1}_{\bigcup_n A_n})(x) \\ &= (T(\sum_n \mathbf{1}_{A_n}))(x) \\ &= (T(\sup_n g_n))(x) \\ &= (\sup_n T(g_n))(x) \\ &= \sup_n \sum_{k \leq n} T(\mathbf{1}_{A_k})(x) \\ &= \sum_n T(\mathbf{1}_{A_n})(x) \\ &= \sum_n P(x, A_n) \end{aligned}$$

using the sequential normality of T .

$T = T_P$: for any $A \in \Sigma$ and $x \in X$, we have

$$(T_P\mathbf{1}_A)(x) = \int_Y \mathbf{1}_A(y) P(x, dy) = \int_A P(x, dy) = P(x, A) = (T\mathbf{1}_A)(x).$$

By linearity, we get that $T = T_P$ on simple functions.

For $g \in \mathcal{B}(Y, \Sigma)^+$, we know from measure theory that every positive measurable function is the pointwise limit of an increasing sequence of simple functions $g_n = \sum_{k=1}^{N_n} c_k \mathbf{1}_{A_k} \in \mathcal{B}(Y, \Sigma)^+$ with $g(x) = \sup_n g_n(x)$ [10]. Hence

$$Tg = T \sup_n g_n = \sup_n Tg_n = \sup_n T_P g_n = T_P \sup_n g_n = T_P g$$

using the sequential normality of T and T_P .

Now for arbitrary functions in $\mathcal{B}(Y, \Sigma)$, we decompose in real/imaginary part and then positive/negative part and conclude by linearity. $\square$

We will now extend our definition of Markov operators (on bounded measurable functions) to essentially bounded functions. If μ is a measure on $(X, \mathcal{A})$, we will denote $L^p(X, \mathcal{A}, \mu)$ by $L^p(\mu)$, for $p \in [1, \infty]$. Recall that elements of L^p spaces are equivalence classes of almost everywhere equal functions. Hence a function $f \in L^p(\mu)$ is positive if and only if $f(x) \geq 0$ for μ -almost every $x \in X$.

Definition 2.20 (Markov operator on L^∞). Let (X, Ω, M) and (Y, Σ, m) be measure spaces. Any linear operator $T : L^\infty(Y, \Sigma, m) \rightarrow L^\infty(X, \Omega, M)$ that is positive and unital is called a *Markov operator* (on L^∞).

Let (X, Ω, M) and (Y, Σ, m) be σ -finite measure spaces and let $P : X \times \Sigma \rightarrow [0, 1]$ be a Markov kernel from X to Y with the extra assumption that

$$A \in \Sigma \text{ and } m(A) = 0 \implies p_x(A) := P(x, A) = 0 \text{ for } M\text{-a.e. } x. \quad (1)$$

We immediately have that $g \in L^\infty(m) \implies g \in L^\infty(p_x)$ and $\|g\|_{L^\infty(p_x)} \leq \|g\|_{L^\infty(m)}$ whenever $g \in L^\infty(m)$. Moreover, g is thus integrable with respect to the finite measure p_x for M -a.e. $x \in X$ and

$$\left| \int_Y g(y) p_x(dy) \right| \leq \int_Y |g(y)| p_x(dy) \leq \|g\|_{L^\infty(p_x)} \leq \|g\|_{L^\infty(m)}$$

for M -a.e. x . Hence, we can define $T : L^\infty(Y, \Sigma, m) \rightarrow L^\infty(X, \Omega, M)$ by

$$(Tg)(x) := \int_Y g(y) p_x(dy).$$

Essentially as seen previously, T is positive and unital, making it a Markov operator on L^∞ . We briefly recall the setting of σ -weak topologies, following [8], to study some continuity and duality properties of T .

To any Banach space X , i.e. a complete normed vector space over $\mathbb{K}$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, one can associate the Banach space of all bounded linear maps from X to $\mathbb{K}$ called the dual space of X , denoted X^* . The weak* topology on X^* is the topology on X^* induced by the family of seminorms $\{p_x\}_{x \in X}$ given by

$$p_x(\varphi) = |\varphi(x)|, \quad \forall x \in X, \varphi \in X^*.$$

It is most easily described as follows : a net $\{\varphi_\alpha\}$ in X^* converges to $\varphi \in X^*$ in the weak* topology if and only if

$$\varphi_\alpha(x) \xrightarrow{\alpha} \varphi(x), \quad \forall x \in X.$$

By Riesz's theorem [5], for any bounded linear functional $\varphi \in (L^1(M))^*$, there is a unique $g \in L^\infty(M)$ such that $\varphi(f) = \varphi_g(f) = \int_X f(x)g(x)dM(x)$. Using this isomorphism, we will identify $(L^1(X, \Omega, M))^*$ with $L^\infty(X, \Omega, M)$. Then the weak* topology on $L^\infty(X, \Omega, M)$ is sometimes called the σ -weak topology and a net $\{f_\alpha\}$ in $L^\infty(M)$ converges to $f \in L^\infty(M)$ σ -weakly if and only

$$\int_X f_\alpha(x)g(x)dM(x) \xrightarrow{\alpha} \int_X f(x)g(x)dM(x), \quad \forall g \in L^1(M).$$

Proposition 2.21. *T is continuous for the σ -weak topologies on $L^\infty(Y, \Sigma, m)$ and $L^\infty(X, \Omega, M)$.*

Proof. Let $\{g_\alpha\}_{\alpha \in A}$ be a net in $L^\infty(m)$ converging σ -weakly to $g \in L^\infty(m)$ i.e.

$$\int_Y g_\alpha(y)f(y)dm(y) \xrightarrow{\alpha} \int_Y g(y)f(y)dm(y),$$

for all $f \in L^1(m)$.

We need to show that $\{Tg_\alpha\}_{\alpha \in A}$ converges σ -weakly to Tg in $L^\infty(M)$.

Let $h \in L^1(M)$ and, for M -a.e. $x \in X$, write $\frac{dp_x}{dm}$ for the Radon-Nikodym derivative of p_x w.r.t. m . We have

$$\begin{aligned} \int_Y \left| \int_X h(x) \frac{dp_x}{dm}(y) dM(x) \right| dm(y) &\leq \int_Y \int_X \left| h(x) \frac{dp_x}{dm}(y) \right| dM(x) dm(y) \\ &= \int_X |h(x)| \left(\int_Y \frac{dp_x}{dm}(y) dm(y) \right) dM(x) \\ &= \int_X |h(x)| p_x(Y) dM(x) \\ &= \|h\|_{L^1(M)} < \infty, \end{aligned}$$

by the definition of the Radon-Nikodym derivative and Tonelli's theorem. We can then use Fubini's theorem to obtain

$$\begin{aligned} \int_X (Tg_\alpha)(x)h(x)dM(x) &= \int_X \left(\int_Y g_\alpha(y)dp_x(y) \right) h(x)dM(x) \\ &= \int_X \int_Y h(x)g_\alpha(y) \frac{dp_x}{dm}(y) dm(y) dM(x) \\ &= \int_Y g_\alpha(y) \left(\int_X h(x) \frac{dp_x}{dm}(y) dM(x) \right) dm(y). \end{aligned}$$

As $\int_X h(x) \frac{dp_x}{dm}(y) dM(x) \in L^1(m)$, the σ -weak convergence of $\{g_\alpha\}$ gives us

$$\begin{aligned} \int_Y g_\alpha(y) \left(\int_X h(x) \frac{dp_x}{dm}(y) dM(x) \right) dm(y) &\xrightarrow{\alpha} \int_Y g(y) \left(\int_X h(x) \frac{dp_x}{dm}(y) dM(x) \right) dm(y) \\ &= \int_X (Tg)(x)h(x)dM(x), \end{aligned}$$

from the same computations as above.

Hence, $Tg_\alpha \rightarrow Tg$ σ -weakly. This shows that T is continuous for the σ -weak topologies on $L^\infty(M)$ and $L^\infty(m)$ $\square$

As T is completely positive by an argument similar to the one in Proposition 2.17, we have, by Proposition III.2.2.2 in [4] that T is normal, i.e. the property of sequential normality extends to arbitrary nets of functions, not only sequences.

We get therefore the following result : assume $T : L^\infty(Y, \Sigma, m) \rightarrow L^\infty(X, \Omega, M)$ is linear, positive, unital and normal. Then there exist a Markov kernel $P : X \times \Sigma \rightarrow [0, 1]$ satisfying the absolute continuity condition (1) such that T is the Markov operator associated to P . Indeed by defining $P : X \times \Sigma \rightarrow [0, 1]$ by $P(x, A) = (T\mathbf{1}_A)(x)$ we have

$$A \in \Sigma \text{ and } m(A) = 0 \implies \mathbf{1}_A(y) = 0, m\text{-a.e.} \implies P(x, A) = (T\mathbf{1}_A)(x) = \mathbf{0}_X(x) = 0, M\text{-a.e.}$$

Moreover, by Proposition 2.4.12 in [8], there exists a bounded linear operator $S : L^1(X, \Omega, M) \rightarrow L^1(Y, \Sigma, m)$ such that $S^* = T$. We will characterize this operator S to highlight our parallel with quantum channels. For this, we need a third definition of Markov operators :

Definition 2.22 (Markov operators on densities). Let (X, Ω, M) and (Y, Σ, m) be σ -finite measure spaces and

$$D_X = \{f \in L^1(X, \Omega, M) : f \geq 0, \|f\|_1 = 1\}$$

the subset of L^1 containing all densities.

A linear mapping $S : L^1(X, \Omega, M) \rightarrow L^1(Y, \Sigma, m)$ is called a *Markov operator* (on densities) if $S(D_X) \subset D_Y$.

For every $f \in D_X$, one may associate the map $\mu_f : \Sigma \rightarrow [0, +\infty)$ defined by

$$\mu_f(A) = \int_X f(x)p_x(A)M(dx).$$

μ_f defines a finite measure on (Y, Σ) (σ -additivity of μ_f comes from the positivity of f and σ -additivity of p_x) and moreover it is absolutely continuous w.r.t. m .

Write $Sf := \frac{d\mu_f}{dm}$, the Radon-Nikodym derivative of μ_f w.r.t. m . Then, by properties of the Radon-Nikodym derivative, Sf is a Σ -measurable and positive function. Moreover,

$$\|Sf\|_1 = \int_Y |Sf| dm = \int_Y \frac{d\mu_f}{dm} dm = \mu_f(Y) = \|f\|_1 = 1.$$

Hence, $S(D_X) \subset D_Y$. Note that,

$$S(cf + g) = \frac{d\mu_{cf+g}}{dm} = \frac{d(c\mu_f + \mu_g)}{dm} = c\frac{d\mu_f}{dm} + \frac{d\mu_g}{dm} = cSf + Sg,$$

by linearity of the Radon-Nikodym derivative, for any $c \in \mathbb{R}_+$ and $f, g \in D_X$.

Therefore, we can extend S to a Markov operator on densities by writing any $f \in L^1(X, \Omega, M)$ as $f = \text{Re}(f)^+ - \text{Re}(f)^- + i(\text{Im}(f)^+ - \text{Im}(f)^-)$ and then multiplying and dividing each term by their respective L^1 -norm when it is non-zero, or, omitting the term in the decomposition when it is zero.

We also note that we get

$$\int_Y Sf dm = \int_X f dM, \quad \forall f \in L^1(X, \Omega, M). \quad (2)$$

Proposition 2.23. T is the Banach space adjoint of S , i.e. $S^* = T$.

Proof. We abbreviate $L^1(X, \Omega, M)$ as $L^1(X)$, $L^\infty(Y, \Sigma, m)$ as $L^\infty(Y)$ and so on.

The Banach space adjoint of S is the unique linear map $S^* : (L^1(Y))^* \rightarrow (L^1(X))^*$ satisfying the following property :

$$(S^*\varphi)(f) = \varphi(Sf) \quad (3)$$

$\forall f \in L^1(X), \forall \varphi \in (L^1(Y))^*$.

As mentioned above, we identify $(L^1(Y))^*$ with $L^\infty(Y)$ (and $(L^1(X))^*$ with $L^\infty(X)$) so that S^* is a map from $L^\infty(Y)$ to $L^\infty(X)$. Equation (3) is then equivalent to :

$$\varphi_{S^*g}(f) = \int_X f(x)(S^*g)(x)M(dx) = \int_Y (Sf)(y)g(y)m(dy) = \varphi_g(Sf),$$

for any $f \in L^1(X), g \in L^\infty(Y)$.

Thus, to show that $S^* = T$, we have to check that

$$\varphi_{Tg}(f) = \varphi_g(Sf), \quad (4)$$

for all $f \in L^1(X)$ and $g \in L^\infty(Y)$.

Let $A \in \Sigma$ and $f \in L^1(X)$. Then we have

$$\varphi_{\mathbf{1}_A}(Sf) = \int_Y (Sf)(y)\mathbf{1}_A(y)m(dy) = \int_A (Sf)(y)m(dy) = \mu_f(A),$$

by definition of the Radon-Nikodym derivative, and

$$\varphi_{T\mathbf{1}_A}(f) = \int_X f(x)(T\mathbf{1}_A)(x)M(dx) = \int_X f(x)p_x(A)M(dx) = \mu_f(A),$$

by definition of μ_f .

By linearity, Equation (4) holds for all $f \in L^1(X)$ and all Σ -measurable simple functions on Y .

Now for $g \in L^\infty(Y)^+$, approximate by an increasing sequence of positive simple functions $\{s_n\}$ s.t. $g(y) = \sup_n s_n(y) = \lim_{n \rightarrow \infty} s_n(y)$. For any $h \in L^1(Y)$, we have $\lim_{n \rightarrow \infty} h(y)s_n(y) = h(y)g(y)$ and $|h(y)s_n(y)| \leq |h(y)|\sup_n s_n(y) = |h(y)|g(y)$ which is integrable. Hence, by the Dominated Convergence Theorem, $s_n \rightarrow g$ σ -weakly and

$$\begin{aligned} \varphi_g(Sf) &= \int_Y (Sf)(y)g(y)m(dy) \\ &= \int_Y \lim_{n \rightarrow \infty} (Sf)(y)s_n(y)m(dy) \\ &= \lim_{n \rightarrow \infty} \int_Y (Sf)(y)s_n(y)m(dy) \\ &= \lim_{n \rightarrow \infty} \varphi_{s_n}(Sf) \\ &= \lim_{n \rightarrow \infty} \varphi_{Ts_n}(f) \end{aligned}$$

Moreover, with similar arguments, we use the dominated convergence theorem again and the σ -weak continuity of T to obtain :

$$\lim_{n \rightarrow \infty} \varphi_{Ts_n}(f) = \varphi \lim_{n \rightarrow \infty} Ts_n(f) = \varphi_{Tg}(f)$$

For an arbitrary $g \in L^\infty(Y)$, decompose in real/imaginary and positive/negative part. Consequently, T also satisfies Equation (3) $\implies S^* = T$. $\square$

2.4 Review

To summarize and transition to the non-commutative setting, we quickly review our construction. Recall that any Markov kernel $P : X \times \Sigma \rightarrow [0, 1]$ from X to Y , i.e. a

family of probability measures $\{p_x\}_{x \in X}$ defines a linear operator $(Tg)(x) = \int g \, dp_x$ on bounded measurable functions. It is then unital, positive and sequentially normal, and all such maps are of this form.

Moreover, if we extend these operators to essentially bounded functions on a measure space, we have, with the extra assumption of the underlying kernel being almost everywhere absolutely continuous, that they are the adjoints of positive, norm-preserving, linear maps on densities.

Hence, we have found a pair $T : L^\infty(Y) \rightarrow L^\infty(X)$, $S : L^1(X) \rightarrow L^1(Y)$ with $S^* = T$ such that S is positive, "trace"-preserving (Equation (2)), T is unital, completely positive and adding the normality allows us to create an integral representation via Markov kernels.

The spaces $L^\infty(X)$ and $L^\infty(Y)$ are examples of commutative von Neumann algebras with preduals $L^1(X)$ and $L^1(Y)$ respectively. In this setting, complete positivity is actually granted from positivity itself [9]. However, in the non-commutative setting of bounded operators on a Hilbert space, this is not the case and complete positivity attains its full sense. We will now study this non-abelian framework to highlight our analogy.

3 Quantum Channels

In this section, we will work in the noncommutative setting $\mathcal{B}(\mathcal{H})$ of bounded linear operators on an infinite dimensional separable complex Hilbert space $\mathcal{H}$ following the work of S. Attal in [2] and R. Rebolledo in [9].

3.1 Completely Positive Maps

To motivate the definition of quantum channels, we look again at Definition 2.13 of complete positivity in the case where $\mathfrak{A} = \mathcal{B}(\mathcal{H})$ and $\mathfrak{B} = \mathcal{B}(\mathcal{K})$ for Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$.

As showed in [3] or [4], $\mathcal{B}(\mathcal{H})$ endowed with the operator norm is a C^* -algebra for the involution given by the Hilbert space adjoint with predual

$$\mathcal{T}(\mathcal{H}) = \{T : \mathcal{H} \rightarrow \mathcal{H} \mid T \text{ linear, bounded and } \text{Tr}[|T|] < \infty\}$$

the space of *trace-class* operators with norm $\|T\|_1 = \text{Tr}[|T|]$.

Now, for each $n \geq 1$, let $\mathcal{M}_n(\mathcal{B}(\mathcal{H}))$ denote the C^* -algebra of $n \times n$ -matrices $[A_{ij}]$, with $A_{ij} \in \mathcal{B}(\mathcal{H})$. For any linear map $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{K})$, we associate the map

$$\begin{aligned} \Phi_n : \mathcal{M}_n(\mathcal{B}(\mathcal{H})) &\rightarrow \mathcal{M}_n(\mathcal{B}(\mathcal{K})) \\ \Phi_n([A_{ij}]) &\mapsto [\Phi(A_{ij})]. \end{aligned}$$

Proposition 3.1. *Given Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$, a linear map $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{K})$ is completely positive if and only if $\Phi_n : \mathcal{M}_n(\mathcal{B}(\mathcal{H})) \rightarrow \mathcal{M}_n(\mathcal{B}(\mathcal{K}))$ is positive for all $n \geq 1$.*

Proof. Let $A_i \in \mathcal{B}(\mathcal{H})$ and $B_i \in \mathcal{B}(\mathcal{K})$ for $i = 1, \dots, n$. For $u \in \mathcal{K}$, we have

$$\left\langle u, \sum_{i,j=1}^n B_i^* \Phi(A_i^* A_j) B_j u \right\rangle = \left\langle \begin{bmatrix} B_1 u \\ \vdots \\ B_n u \end{bmatrix}, \begin{bmatrix} \Phi(A_1^* A_1) & \cdots & \Phi(A_1^* A_n) \\ \vdots & \ddots & \vdots \\ \Phi(A_n^* A_1) & \cdots & \Phi(A_n^* A_n) \end{bmatrix} \begin{bmatrix} B_1 u \\ \vdots \\ B_n u \end{bmatrix} \right\rangle = \langle [B_i u], \Phi_n([A_i^* A_j]) [B_i u] \rangle.$$

Hence, the proposition follows since $[A_i^* A_j]$ is positive. $\square$

Consider the n -dimensional Hilbert space $\mathbb{C}^n$ and a Hilbert space $\mathcal{H}$. The space $\mathcal{B}(\mathcal{H} \otimes \mathbb{C}^n) \cong \mathcal{B}(\mathcal{H}) \otimes \mathcal{B}(\mathbb{C}^n)$ can be identified with $\mathcal{M}_n(\mathcal{B}(\mathcal{H}))$ by mapping a matrix $[A_{ij}]$ to $\sum_{i,j} A_{ij} \otimes |i\rangle \langle j|$, where $\{|i\rangle\}_{i=1}^n$ is the computational basis of $\mathbb{C}^n$ (inverse map is $A \otimes B \mapsto [Ab_{ij}]$). Hence the map Φ_n can be seen as follows :

$$\begin{aligned}\Phi_n : \mathcal{B}(\mathcal{H}) \otimes B(\mathbb{C}^n) &\rightarrow \mathcal{B}(\mathcal{K}) \otimes B(\mathbb{C}^n) \\ A \otimes B &\mapsto \Phi(A) \otimes B\end{aligned}$$

We will now look at the definition of quantum channels and give an example to show why this characterization of complete positivity is relevant.

3.2 Open Quantum Systems and Quantum Channels

In quantum theory, quantum channels are the maps that represent the most general "physically admissible" transformation a quantum system can undergo. We have a few physical considerations to take into account before characterizing these maps :

- in practice it is very difficult to isolate a quantum system from its environment. Hence one has to consider quantum systems that interact with other ones. These are the so-called *open quantum systems*.
- Moreover, in general one does not have access to the second system which could be too large, complicated, like a large environment, a quantum heat bath, etc.

In this open system formalism, the quantum states of a quantum system with state space a Hilbert space $\mathcal{H}$ are the trace-class, positive operators $\rho : \mathcal{H} \rightarrow \mathcal{H}$ such that $\text{Tr } \rho = 1$. This is the Schrödinger picture of quantum mechanics.

Quantum mechanics being inherently linear, we expect any sensible transformation to be a linear map that sends states of a system A to states of a transformed system B i.e. $\Phi : \mathcal{T}(\mathcal{H}_A) \rightarrow \mathcal{T}(\mathcal{H}_B)$ should be :

1. linear
2. positive
3. trace preserving, i.e. $\text{Tr}[\Phi(X)] = \text{Tr } X$, for all $X \in \mathcal{T}(\mathcal{H}_A)$.

Those three conditions allow us to show that Φ is bounded in the trace norm. Indeed, for $X \in \mathcal{T}(\mathcal{H}_A)$ self-adjoint, Proposition 1.74 in [1] allows us to write $X = X^+ - X^-$ for X^+, X^- bounded, positive and $|X| = X^+ + X^-$. Then $\|X\|_1 = \text{Tr } |X| = \text{Tr } X^+ + \text{Tr } X^-$ and, because Φ is positive, $\Phi(X^+)$ and $\Phi(X^-)$ are positive. Therefore,

$$\|\Phi(X)\|_1 = \|\Phi(X^+) - \Phi(X^-)\|_1 \leq \text{Tr } \Phi(X^+) + \text{Tr } \Phi(X^-) = \text{Tr } X^+ + \text{Tr } X^- = \|X\|_1$$

. For arbitrary $X \in \mathcal{T}(\mathcal{H}_A)$, write

$$X = A + iB, \quad A = \frac{X + X^*}{2}, \quad B = \frac{X - X^*}{2i},$$

with A, B self-adjoint. By Theorem 2.8 in [3], $X^* \in \mathcal{T}(\mathcal{H}_A)$ so $A, B \in \mathcal{T}(\mathcal{H}_A)$ and

$$\|\Phi(X)\|_1 \leq \|\Phi(A)\|_1 + \|\Phi(B)\|_1 \leq \|A\|_1 + \|B\|_1 \leq 2\|X\|_1.$$

This boundedness allows us to consider the dual map Φ^* acting on the elements of $\mathcal{B}(\mathcal{H}_B)$, called the observables of system B. This setting is called the Heisenberg picture of quantum mechanics. $\Phi^* : \mathcal{B}(\mathcal{H}_B) \rightarrow \mathcal{B}(\mathcal{H}_A)$ is uniquely characterized by the identity :

$$\langle \Phi(X), Y \rangle = \langle X, \Phi^*(Y) \rangle \quad (5)$$

for all $X \in \mathcal{T}(\mathcal{H}_A)$, $Y \in \mathcal{B}(\mathcal{H}_B)$, with the duality pairing $\langle \rho, A \rangle = \text{Tr}[\rho A]$. To see if conditions 1., 2. and 3. are enough to characterize quantum channels, we will translate them in this dual picture, i.e. Φ^* should be :

- a. linear
- b. positive
- c. unital, i.e. $\Phi^*(I_B) = I_A$.

Unfortunately, condition b. is slightly too weak to describe the physically reasonable transform of open quantum systems. We will use an example from [2, Section 6.3.1] to explain why.

Consider a linear map $\mathcal{N} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ such that $\mathcal{N}(A) \in \mathcal{B}(\mathcal{H})^+$ for all $A \in \mathcal{B}(\mathcal{H})^+$. Now imagine that outside the system represented by $\mathcal{H}$ is another quantum system represented by $\mathcal{K}$, which is independent of $\mathcal{H}$, in the sense that the two quantum systems have no interaction whatsoever. Consider the whole system $\mathcal{H} \otimes \mathcal{K}$ and the transformation $\widehat{\mathcal{N}}$ which consists in applying the transformation $\mathcal{N}$ to $\mathcal{H}$ and ignoring $\mathcal{K}$. This means considering the mapping

$$\begin{aligned} \widehat{\mathcal{N}} &= \mathcal{N} \otimes I : \mathcal{B}(\mathcal{H} \otimes \mathcal{K}) \rightarrow \mathcal{B}(\mathcal{H} \otimes \mathcal{K}) \\ A \otimes B &\mapsto \mathcal{N}(A) \otimes B. \end{aligned}$$

Then this mapping might not preserve positivity anymore and its predual might not map quantum states to quantum states.

Let us show a counter-example. Take $\mathcal{H} = \mathcal{K} = \mathbb{C}^2$ and $\mathcal{N}$ to be the transpose mapping on $\mathcal{B}(\mathcal{H})$, namely, $\mathcal{N}(A) = A^t = \overline{A}^*$. We can see $\mathcal{B}(\mathcal{H} \otimes \mathcal{K})$ as the algebra of 2×2 matrices with coefficients in $\mathcal{B}(\mathcal{H})$ and the mapping $\widehat{\mathcal{N}}$ acts as follows :

$$\widehat{\mathcal{N}} : \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \mapsto \begin{bmatrix} \mathcal{N}(A_{00}) & \mathcal{N}(A_{01}) \\ \mathcal{N}(A_{10}) & \mathcal{N}(A_{11}) \end{bmatrix}.$$

Hence, the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \text{ is mapped to } \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The first matrix is positive (it is the rank-one projection onto $|00\rangle + |11\rangle$) whereas the second is not (eigenvalues are +1 and -1), therefore the mapping $\widehat{\mathcal{N}}$ is not positivity preserving.

It is clearly physically unsatisfactory to consider quantum state transforms which are not quantum states transforms when considered as part of a larger world, even if there is no interaction with the environment.

Hence, we ask the map Φ^* defined in Equation (5) to satisfy that, for any $n \in \mathbb{N}$, the natural ampliation

$$\begin{aligned}\Phi_n^* : \mathcal{B}(\mathcal{H}_B) \otimes \mathcal{B}(\mathbb{C}^n) &\rightarrow \mathcal{B}(\mathcal{H}_A) \otimes \mathcal{B}(\mathbb{C}^n) \\ X \otimes Y &\mapsto \Phi^*(X) \otimes Y\end{aligned}$$

is positivity-preserving.

We have seen in Proposition 3.1 that this is exactly equivalent to Φ^* being completely positive. Moreover, by Proposition 2.4.12 in [8], Φ^* is σ -weak continuous which is equivalent to normality by Proposition III.2.2.2 in [4]. Thus, we arrive to :

Definition 3.2. A *quantum channel* (in the Heisenberg picture) from $\mathcal{B}(\mathcal{H}_B)$ to $\mathcal{B}(\mathcal{H}_A)$ is a linear map $\mathcal{N} : \mathcal{B}(\mathcal{H}_B) \rightarrow \mathcal{B}(\mathcal{H}_A)$ which is normal, completely positive and unital.

A *quantum channel* (in the Schrödinger picture) from $\mathcal{T}(\mathcal{H}_A)$ to $\mathcal{T}(\mathcal{H}_B)$ is a bounded linear map $\Phi : \mathcal{T}(\mathcal{H}_A) \rightarrow \mathcal{T}(\mathcal{H}_B)$ such that its dual is a quantum channel from $\mathcal{B}(\mathcal{H}_B)$ to $\mathcal{B}(\mathcal{H}_A)$.

For characterization of quantum channels in the Schrödinger picture, see [2, Section 6.1].

4 Kraus Representations and the Classical Limit

We now give a theorem by Kraus that will allow us to create a representation of unital completely positive maps that relates to the integral representation of Markov operators via Markov kernels.

Theorem 4.1 (Kraus). *Let $\mathcal{H}$ and $\mathcal{K}$ be complex separable Hilbert spaces. Then a linear map $\mathcal{M} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{K})$ is normal and completely positive if and only if there exists a sequence $\{M_j\}_{j \in \mathbb{N}}$ of bounded linear operators from $\mathcal{K}$ to $\mathcal{H}$ such that the series $\sum_{j=1}^{\infty} M_j^* X M_j$ is strongly convergent for any $X \in \mathcal{B}(\mathcal{H})$ and*

$$\mathcal{M}(X) = \sum_{j=1}^{\infty} M_j^* X M_j, \quad \forall X \in \mathcal{B}(\mathcal{H}).$$

Moreover, $\mathcal{M}$ is unital if and only if

$$\mathcal{M}(I_{\mathcal{H}}) = \sum_{j=1}^{\infty} M_j^* M_j = I_{\mathcal{K}}$$

See [9] for the proof.

We now circle back to our very first definition, the one of Markov kernels, to give a final concrete example (adapted from exercise 29.10 in [7]) of our analogy between Kraus representation of quantum channels and integral representation of Markov operators.

Example 4.2. Let $\mathcal{H}$ be an infinite dimensional separable Hilbert space with orthonormal basis $\{|e_i\rangle\}_{i \in \mathbb{N}}$ and $\{|u_i\rangle\}_{i \in \mathbb{N}}$ a sequence of unit vectors in $\mathcal{H}$.

Define the Kraus operators $K_i : \mathcal{H} \rightarrow \mathcal{H}$ by $K_i = |u_i\rangle \langle e_i|$, i.e. $K_i x = \langle e_i, x \rangle |u_i\rangle$ for all $x \in \mathcal{H}$. It can be shown that $\sum_{i \in \mathbb{N}} K_i^* X K_i$ is strongly convergent for any $X \in \mathcal{B}(\mathcal{H})$ and, hence the map $\mathcal{M} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ defined by $\mathcal{M}(X) = \sum_{i \in \mathbb{N}} K_i^* X K_i$ is a quantum channel. Indeed by Theorem 4.1, it is normal, completely positive and

$$\sum_{i=1}^N K_i^* K_i = \sum_{i=1}^N \langle u_i, u_i \rangle |e_i\rangle \langle e_i| = \sum_{i=1}^N |e_i\rangle \langle e_i| \xrightarrow{N \rightarrow \infty} I_{\mathcal{H}} \quad \text{strongly}$$

hence unital. Recall that diagonal operators of $\mathcal{H}$ (with respect to $\{e_j\}$) are the operators of the form $X = \sum_{j \in \mathbb{N}} x_j |e_j\rangle \langle e_j|$ for a bounded sequence $\{x_j\} \subset \ell^\infty(\mathbb{N})$. For such an operator X , we have

$$\mathcal{M}(X) = \sum_{i=1}^{\infty} K_i^* X K_i = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} x_j \langle u_i, e_j \rangle \langle e_j, u_i \rangle |e_i\rangle \langle e_i| = \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} x_j |\langle u_i, e_j \rangle|^2 \right) |e_i\rangle \langle e_i|$$

which is another diagonal operator with coefficients $y_i = \sum_{j \in \mathbb{N}} x_j p_{ij}$ and $p_{ij} = |\langle u_i, e_j \rangle|^2$. For a fixed $i \in \mathbb{N}$,

$$p_{ij} \geq 0, \forall j \in \mathbb{N} \text{ and } \sum_{j \in \mathbb{N}} p_{ij} = \sum_{j \in \mathbb{N}} |\langle u_i, e_j \rangle|^2 = \|u_i\|^2 = 1,$$

by Parseval's identity (Theorem 26.19 in [10]). Hence the p_{ij} 's form an infinite countable row stochastic matrix and, as in Section 2.2, the map $P : \mathbb{N} \times 2^{\mathbb{N}} \rightarrow [0, 1]$ defined by $P(i, \{j\}) = p_{ij}$ is a Markov kernel.

Moreover, if we identify the diagonal operators of $\mathcal{H}$ with $\ell^\infty(\mathbb{N})$ (by defining $f(j) = x_j$), we can view the restriction of the quantum channel $\mathcal{M}$ to diagonal operators as a map $\tilde{\mathcal{M}} : \ell^\infty(\mathbb{N}) \rightarrow \ell^\infty(\mathbb{N})$ with

$$(\tilde{\mathcal{M}}f)(i) = \sum_{j=1}^{\infty} f(j) p_{ij}, \quad \forall i \in \mathbb{N}, \forall f \in \ell^\infty(\mathbb{N}).$$

To recover Markov operators, we look at $\ell^\infty(\mathbb{N})$ from another point of view.

By Example 13.11 in [10], we have that $L^\infty(\mathbb{N}, 2^{\mathbb{N}}, \mu_{\text{count}}) = \ell^\infty(\mathbb{N})$, where $\mu_{\text{count}}(A) = |A|$ if $A \subset 2^{\mathbb{N}}$ is finite and $+\infty$ otherwise. Moreover, $\int_{\mathbb{N}} u \, d\mu = \sum_{n \in \mathbb{N}} u(n)$.

Now, for $P : \mathbb{N} \times 2^{\mathbb{N}} \rightarrow [0, 1]$ defined as above, it stands that

$$\mu_{\text{count}}(A) = 0 \iff A = \emptyset \implies P_i(A) = 0, \forall i \in \mathbb{N}, \forall A \subset \mathbb{N},$$

meaning that $P_i \ll \mu_{\text{count}}$ and, by a.e.-uniqueness of the Radon-Nikodym derivative, that $\frac{dP_i}{d\mu_{\text{count}}}(j) = p_{ij} = P(i, \{j\})$.

Then, for $T_P : L^\infty(\mathbb{N}, 2^{\mathbb{N}}, \mu_{\text{count}}) \rightarrow L^\infty(\mathbb{N}, 2^{\mathbb{N}}, \mu_{\text{count}})$ the Markov operator with integral representation given by the Markov kernel P , we have,

$$(T_P f)(i) = \int_{\mathbb{N}} f(j) dP_i(j) = \int_{\mathbb{N}} f(j) \frac{dP_i}{d\mu_{\text{count}}}(j) d\mu_{\text{count}}(j) = \sum_{j=1}^{\infty} f(j) p_{ij} = (\tilde{\mathcal{M}}f)(i),$$

for all $i \in \mathbb{N}$ and $f \in \ell^\infty(\mathbb{N})$.

Hence, the restriction of the quantum channel $\mathcal{M}$ to diagonal operators in $\mathcal{B}(H)$ is actually just a Markov operator on $L^\infty(\mathbb{N})$, where the probabilities $p_{ij} = |\langle u_i, e_j \rangle|^2$ of the corresponding Markov kernel can be interpreted, in quantum mechanics, as the probability of outcome e_j when performing a measurement in the basis $\{|e_j\rangle\}$ given that the system is in the state $|u_i\rangle$.

5 Conclusion

In this project, we studied quantum channels as noncommutative analogues of classical Markov operators. Starting from Markov kernels on measurable spaces, we constructed positive unital operators through integral representations and analysed their duality and

continuity properties on L^∞ and L^1 spaces. We then introduced quantum channels on $\mathcal{B}(\mathcal{H})$ as normal completely positive unital maps and related their Kraus representations to the integral representations of Markov operators.

Moreover, by restricting suitable quantum channels to diagonal operators, we recovered classical Markov operators associated with stochastic matrices and Markov kernels, illustrating how classical stochastic dynamics arise as commutative instances of quantum probabilistic structures.

The framework developed in Section 3.2 naturally extends beyond $\mathcal{B}(\mathcal{H})$ to arbitrary von Neumann algebras. In particular, commutative von Neumann algebras of the form $L^\infty(X)$ recover the classical theory of normal Markov operators. This highlights the role of quantum channels as a general operator-algebraic framework encompassing both classical and quantum stochastic dynamics.

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